



DPULookUpSequence

Retrieves for an input value the corresponding output value from a list of values.

1. Inputs:

In Input value

2. Parameters:

Sequence List of elements (input window, output value). See below for a detailed description.

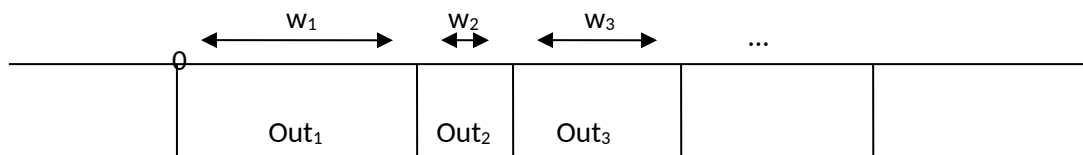
3. Outputs:

Out Lookup value

SequenceWindow With of current input window

4. Functional principle:

DPULookUpSequence subdivides the value range of the input value into s.c. input windows w_1 , w_2 , For each input window the corresponding (constant) output value Out_1 , Out_2 , ... can be defined:



The width of each input window as well as the corresponding output value is defined in the list *Sequence*:

$Sequence = (w_1, Out_1, w_2, Out_2, w_3, Out_3, \dots);$

The first window always has 0 as its left boundary. If $In < 0$ then *Out* will be set to Out_1 . If *In* is greater than the right boundary of the last window then *Out* will be set to the output value of the last window.

5. Example:

```
DPULookUpSequence Seq {  
    Computer = {XYZ};  
    Index = 10;  
    Sequence = (1, 4,    3, 2,    2, 5);  
};
```



SILAB DRIVING SIMULATION

version Version 7.1 documentation

In this example, the DPU computes the following function:

$$Out \leftarrow \begin{cases} 4 & | \quad In < 1 \\ 2 & | \quad 1 \leq In \leq 4 \\ 5 & | \quad In > 4 \end{cases}$$